

AI Robot Autonomy CEAMLS Summer AI Research Institute

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Abstract— Travel poses many challenges, especially for individuals who face mobility issues. Significant advancements in autonomous and semi-autonomous wheelchairs have been made to aid mobility, aiming to create a fully autonomous wheelchair capable of navigating complex environments like airports. These advanced wheelchairs integrate various technologies to enhance personal independence and safety. Although studies are being proposed for urban settings¹, others have not yet created a program that is capable of full autonomy in bringing an individual to any customized location with obstacles along the path (as one would expect in a busy airport, museum, or mall). For instance, the LUCI system improves semi-autonomous wheelchairs by incorporating sensor-based safety measures that collaborate with user inputs for safer navigation and obstacle avoidance². However, fully autonomous navigation in dynamic environments such as busy airports or malls remains a challenge, both in scope and range of access, including the number of wheelchairs available. Current systems often rely on extensive sensor arrays and mapping technologies but struggle with frequently changing obstacles³. We aim to develop a hybrid system using existing SLAM-based models combined with a track-based model to produce a fully autonomous wheelchair capable of navigating the entirety of BWI airport, from ticketing to the gate, without additional user inputs. While research is being conducted for urban settings, current programs can only autonomously navigate limited areas using customized paths in such dynamic environments. Achieving whole-building, autonomous airport navigation would be a significant advancement in autonomous AI for mobility assistance.

Index Terms— Autonomous Wheelchairs, Computer Vision, Track-Based Movement, SLAM,

Enter keywords or phrases in alphabetical order, separated by commas. For a list of suggested keywords visit http://www.ieee.org/organizations/pubs/ani_prod/keywrd98.txt

INTRODUCTION

Intelligent robots are currently developed to help handicapped people across a plethora of contexts. For instance, intelligent wheelchairs play an important role in helping the handicapped and the elderly navigate environments with which they otherwise might struggle. In this paper, we will illustrate the development of an intelligent wheelchair capable of both track and visual-based navigation in complex environments.

Problem Statement

A gap present in the field of autonomous wheelchairs is that most semi-autonomous wheelchairs are designed for individual use, not a large-scale public format. Moreover, there is currently no track-based autonomous wheelchair on the market. To our knowledge, there is still a lack of programs that are capable of full autonomy in bringing individuals to any customized location within an airport: current models are limited to specific areas and a small number of working models. Navigating through an airport, from ticketing to any gate, through crowds, without additional user inputs will be a significant step forward in the field of autonomous AI.

Motivation

Autonomous wheelchairs have the potential to significantly enhance the independence and mobility of individuals with limited mobility. This technology can improve quality of life by providing reliable and efficient transportation in busy and complex environments, such as airports, railways, and other areas. Developing such advanced autonomous systems presents numerous technical challenges and opportunities for innovation in robotics, AI, and human-computer interaction. Achieving full autonomy in a public, obstacle-rich environment will push the boundaries of current technology and open new avenues for research and application.

I. BACKGROUND

A. The work presented in this paper builds on Simultaneous Localization and Mapping (SLAM) technologies, and integrates a track-based approach onto this model to offer a more versatile and holistic approach to autonomous and semi-autonomous wheelchair technology. Service robots operating in populated environments generally achieve success by utilizing an array of sensors and advanced system designs grounded in statistical methods. For instance, autonomous robots use a common control pipeline, a closed loop system with sensors and feedback⁴. This includes the use of exteroceptive sensors, which sense the immediate external

¹ Nijijima, S., Sasaki, Y., & Mizoguchi, H. (2019). *Real-time autonomous navigation of an electric wheelchair in large-scale urban area with 3D map*

² LUCI enables semi-autonomous robotic wheelchair controls. (2022, December 2). Sunrise Medical. <https://www.sunrisemedical.com/education-in-motion/blog/december-2022/luci-enables-semi-autonomous-wheelchair-controls>

³ Luke, B. M., Tripathi, A., & Sampson, S. A. (2023). Autonomous wheelchair robot for enhanced airport accessibility. *2023 International Conference on Next Generation Electronics (NEleX)*. <https://doi.org/10.1109/nelex59773.2023.10420850>

⁴ Juneja, A., Bhandari, L., Mohammadbagherpoor, H., Singh, A., & Grant, E. (2019). A comparative study of SLAM algorithms for indoor navigation of autonomous wheelchairs. *2019 IEEE International Conference on Cyborg and Bionic Systems (CBS)*.

environment of the robot. Some examples include: LiDAR, Camera, Ultrasonic Sensor, and Radar. It also includes the use of proprioceptive sensors, which sense and measure the internal state of the robot itself; some examples include Encoders and IMU.

B. A separate part of the pipeline deals with localization. The landmarks generated by the perception algorithm are used to learn a map and to estimate the pose, i.e., position and orientation of the robot within the generated 2D map. SLAM condenses the localization and mapping processes into a single algorithm. Introduced in 1991⁵, SLAM addresses the problem of a moving robot in an unknown environment, determining how likely it is to construct a consistent map of its surroundings while estimating its current location and navigating through the area. Some SLAM solutions require a combination of sensors, while others only need one. These algorithms process incoming sensor data to build a map by observing the environment, identifying landmarks, and simultaneously estimating the location of the moving platform relative to those landmarks. Uncertainty in mapping and localization is often due to noisy sensor measurements, which can lead to an inaccurate perception of the environment.

C. There have been airports which incorporate autonomous wheelchairs into their design using computer vision methods similar to SLAM. Miami International Airport was the first U.S. airport to integrate autonomous wheelchairs into their layout and design, starting around June 2024⁶. This program was introduced to aid in the transport of airline passengers with limited mobility to their desired gate through the use of a touch-screen app, but is limited to two terminals and is exclusively used by American Airlines passengers. Similarly, Tokyo's Haneda Airport began to integrate self-driving wheelchairs starting around June 2020⁷. The addition of autonomous wheelchairs in multiple airports illustrates a trend of transitioning to modern technological solutions to help individuals with mobility issues with navigating complex environments. Our project expands on the work of these airports by integrating track-based technologies on top of computer vision methods, with the goal of creating an autonomous wheelchair for use throughout the entirety of BWI airport and available to all passengers, regardless of airline or destination gate.

⁵ Leonard, J. J. & Durrant-Whyte, H. F. "Mobile robot localization by tracking geometric beacons". IEEE Transactions on robotics and Automation 7.3 (1991), pp. 376–382.

⁶ *New autonomous wheelchairs at Miami International Airport*. (2024, June 25). New autonomous wheelchairs at Miami International Airport. <https://news.miami-airport.com/miami-dade-mayor-and-american-airlines-announce-new-autonomous-wheelchairs-at-mia/>

⁷ Chang, B. (2020, June 24). *Tokyo's Haneda airport is adding self-driving electric wheelchairs for social distancing — here's how they work*. Business Insider. <https://www.businessinsider.com/tokyos-haneda-airport-will-autonomous-mobility-service-2020-6>

II. METHODS

The methods used to develop an autonomous wheelchair using SLAM and track-based navigation include deep learning, computer vision, and LiDAR. The experiments were conducted in a controlled indoor environment designed to simulate real-world conditions. The test area included various obstacles, such as furniture and dynamic objects, to evaluate the wheelchair's navigation capabilities. The setup aimed to replicate the complexity of environments like airports.

A. SLAM Prototyping

In this project, SLAM implementation was done on a miniature-scale robot for prototyping. This approach not only alleviated the cost constraints but also provided advantages like easy visualization of ideas, identification of problems at an early stage of development, easy feedback acquisition, and encouraged innovation and creativity in the project.

The SLAM implementation of this project was done in two phases, LiDAR-SLAM implementation and hybrid Visual SLAM with LiDAR-SLAM implementation.

The LiDAR sensor in conjunction with robot odometry was utilized in implementing LiDAR-SLAM to build a 2D map of the environment while navigating autonomously. This means using a distance measurement obtained with high accuracy from the LiDAR sensor and positional data from the odometry system of the robot for mapping and localization.

Construction of Robot

Materials:

- Raspberry Pi 4: acts as the central processing unit
- Waveshare General Driver for Robots: controls the motors and IMU
- DFRobot TT Motor with Encoders: provides motion and feedback on distance and speed
- 2S 18650 LiPo Battery Bank: power source.
- 2S 2A Multi-Cell 2S Type-C to 8.4V Step-Up Boost LiPo Polymer Li-Ion Charger: manages battery charging
- FHL-LD19 DTOF Lidar: environmental scanning and mapping
- Acrylic 2WD Robot Chassis: structural framework.

Preliminary Construction:

The motors were attached firmly to the backside of the construction board with screws, then the wheels were fitted onto the motor shafts to ensure a secure attachment. Wires from the motors were connected to the Waveshare motor control board, hooking them up for both power and control signals.

The Raspberry Pi, serving as the robot's main processing unit, was mounted atop the Waveshare board, setting up the connections for power, data, and control signals. An SD card preinstalled with the Ubuntu 22.04 operating system and ROS2 was then installed into the Raspberry Pi.

Then, the LiDAR sensor was mounted securely on the construction board to allow an unobstructed 360° view. The LiDAR was connected to the Raspberry Pi for data transfer and power supply through appropriate wiring. Lastly the battery pack was connected to provide the robot with necessary power. Care was taken to ensure that all connections were secure and the wiring clean so that it wouldn't interfere with the moving parts of the robot.

This assembly process resulted in a robot that could make use of LiDAR and odometry to run autonomous navigation and mapping. With these components put together, the robot is capable of creating a 2-D map of its environment while localizing itself in that map, hence performing effective autonomous operation.

Localization and Mapping with LiDAR:

Mapping and localization using LiDAR included integration with ROS2 (Robot Operating System 2) on the Linorobot2 platform with RViz2 for visualization and debugging.

First, the robot was guided around a whole room manually using a keyboard controlling its movement. During this process, the LiDAR sensor mounted on the robot took distance measurements to create a detailed point cloud of its environment. As the robot moved across the room, the LiDAR sensor data was incrementally obtained and used to build the map.

Loop closure was achieved when the robot reached the position from which it had started. After the successful completion of loop closure, the map was saved. The saved map is loaded into the system and used for autonomous navigation.

During the self-navigation phase, the robot would navigate the environment autonomously without intervention. RViz2 was used to display the map generated and the pose of the robot in real-time, showing the critical aspects of the mapping and navigation processes in detail. RViz2 visualized the mapping and navigation processes so that proper debugging was done on the map and the working of the robot. This provided an integrated approach toward accurate localization and mapping using LiDAR by incorporating manual mapping, loop closure, and autonomous navigation.

Visual SLAM

Our approach to developing an autonomous wheelchair capable of full navigation within an airport environment involved the use of Visual SLAM. Similar to LiDAR SLAM, this involved using a real sense camera and an inertial measurement unit to build a map of an environment while simultaneously tracking the robot position within that map. The process starts with data acquisition where distinctive points in the images are identified and described. The features are then used to estimate the camera's movement for pose estimation. Using these estimated poses, the map of the

environment is incrementally built and refined, all while continuously updating the robot location within this map. The Linorobot2 platform provided a robust foundation for implementing robot mobility (Linorobot2, n.d.; Hippo5329, n.d.) This allowed us to move the robot around from the Jetson Nano and remotely from our laptops. It also provided techniques that allowed us to configure our waveshare board and calibrate our motors.

Wheelchair Prototype Construction

Using the body of the LiDAR robot, this prototype construction began with the assembly of the robot and integration of the necessary hardware components. The primary processing unit, the Jetson Nano leveraging a Samsung solid state drive, was configured to handle both the RealSense camera and the waveshare board which served as the robot's inertial measurement unit. Initially, we faced challenges with SD card image files, which were overcome by cloning existing working SD and SSD cards. The RealSense camera was securely mounted on the robot to ensure an unobstructed view for environmental scanning. All hardware components, including motors, encoders, and the Waveshare motor control board, were connected and properly configured to work seamlessly with the Jetson Nano. The entire setup was powered by an Anker Prime 20,000mAh Power Bank, ensuring adequate and stable power supply for uninterrupted operation.

Sensor Fusion and Calibration

In the development of our autonomous wheelchair, sensor fusion and calibration played a critical role in ensuring accurate localization and navigation. Sensor fusion involves combining data from multiple sensors to achieve more accurate and reliable information than could be obtained from any single sensor. Our system integrates data from various sensors, including a RealSense camera, an IMU, and LiDAR. Each of these sensors provides unique data that, when combined, enhances the overall perception and understanding of the wheelchair's environment. The RealSense camera captures high-resolution depth information, creating detailed point clouds of the environment, while the IMU provides orientation and motion data, helping track the wheelchair's movement and orientation in real-time. By synchronizing the depth information from the camera with the motion data from the IMU, we achieved more precise pose estimation and improved the accuracy of the visual SLAM process. Data from the LiDAR sensor is fused with the RealSense camera's point cloud data, providing a comprehensive 3D representation of the surroundings. This fusion of LiDAR and camera data allows the wheelchair to navigate complex environments with dynamic obstacles more effectively.

However, we encountered significant challenges related to

sensor noise and its impact on the mapping process. The presence of noise in the data from the sensors often confused the robot, leading to inaccuracies in the map and misinterpretation of the environment. This was particularly problematic when the robot encountered dynamic environments with moving objects and frequent changes. The IMU data, in particular, could introduce noise due to vibrations and sudden movements, which sometimes led to errors in pose estimation. Similarly, the RealSense camera occasionally produced noisy depth data, especially in low-light conditions or when dealing with reflective surfaces, which further complicated the mapping process. These issues with noise sometimes caused the robot to misalign the map or fail to recognize previously visited locations, undermining the reliability of the SLAM algorithm. To resolve the issue of sensor noise, we disabled IMU fusion on the RealSense camera, which significantly reduced the noise in the data. We discovered the RealSense camera had an IMU of its own so disabling IMU fusion made the algorithm rely solely on the wavershare board's IMU. This adjustment allowed the camera to provide clearer and more accurate depth information, improving the overall reliability of the visual SLAM process and ensuring better map alignment and pose estimation.

We used a series of controlled movements and rotations to gather data from all sensors simultaneously, then applied calibration algorithms to compute the relative positions and orientations of the sensors. This step ensured that the data from each sensor could be accurately combined, enhancing the reliability of the sensor fusion process. To account for any shifts or changes in sensor alignment during operation, we implemented dynamic calibration techniques. This involved continuously monitoring the sensor data for discrepancies and applying real-time adjustments to maintain accurate alignment. Dynamic calibration was particularly important in maintaining the accuracy of the SLAM process in a constantly changing environment.

Isaac Ros Visual SLAM

For our autonomous wheelchair project, we employed NVIDIA's Isaac ROS Visual SLAM to enhance our mapping and localization capabilities (NVIDIA Corporation, n.d.). Isaac ROS Visual SLAM is a robust framework designed for real-time simultaneous localization and mapping using camera data. It provides efficient algorithms and tools optimized for NVIDIA hardware, such as the Jetson Nano, ensuring high performance and accuracy.

The integration of Isaac ROS Visual SLAM involved several steps. Initially, we set up the Jetson Nano with the Isaac ROS SDK, which required configuring the software environment and ensuring compatibility with our RealSense camera. Using

the Isaac ROS Visual SLAM nodes, we processed the camera's depth and RGB data to perform real-time localization and mapping. This setup allowed the wheelchair to build and update a map of its environment dynamically while accurately tracking its position within that map.

Isaac ROS Visual SLAM also provided advanced features such as loop closure and pose graph optimization, which were crucial for correcting drift and maintaining map accuracy over time. By leveraging the computational power of the Jetson Nano and the efficiency of the Isaac ROS algorithms, we achieved reliable and precise mapping and localization, essential for navigating complex environments like airports.

Visual SLAM Algorithm

The visual SLAM algorithm involves several key steps. Initially, the RealSense camera captures depth and RGB data, which are then processed by the Isaac ROS Visual SLAM nodes. The wavershare board's IMU provides complementary orientation and motion data, which are fused with the camera data to estimate the robot's pose accurately. Advanced features like loop closure and pose graph optimization ensure that the map remains consistent over time by correcting any drift. These features are essential for recognizing previously visited locations and refining the map, which enhances the reliability of the localization and mapping processes. We were able to view and monitor the accuracy of the algorithm through rviz2. Rviz2 which is short for Ros visualization 2 provides a graphical interface for users to view their robot, sensor data, maps, and more.

Mapping and Localization

Point cloud mapping was a crucial part of the visual SLAM process. Using the RealSense camera and Isaac ROS Visual SLAM, we captured a dense collection of data points representing the room we worked in. These points were processed through the Isaac ROS framework to create a detailed point cloud map, which served as the basis for the robot's navigation and localization tasks. The camera captured depth information, allowing the creation of a three-dimensional representation of the environment. This point cloud data was incrementally updated as the robot moved, providing real-time mapping and enabling the robot to navigate complex environments. The preliminary point cloud map was further refined through continuous data acquisition and processing using Isaac ROS Visual SLAM, resulting in a comprehensive and accurate environmental map.

Utilizing the Isaac ROS Visual SLAM framework, we achieved precise and real-time localization by combining data from the RealSense camera and the IMU. The RealSense camera provided detailed depth and RGB data, while the IMU contributed orientation and motion information. Isaac ROS Visual SLAM fused these data streams to continuously

estimate the wheelchair's pose relative to the environment. By placing the robot and the same origin point that was used to map the room we were successfully able to localize it. This allows us to utilize autonomous navigation and publish goal posts.

A2. Track-Based Method

Software Tools

- RoboFlow – Data Annotation and Labeling
- Colab – Model training
- VS Code – Coding and Debugging
- Libraries: Ultralytics, OpenCV, edge-tpu-silva
- Programming Language: Python

Hardware Tools

- Raspberry Pi
- Coral TPU
- Camera
- Motors, motor encoders, motor drivers
- Wheelchair

The track-based method utilized deep learning and computer vision techniques to achieve full autonomy for the wheelchair. A dataset of 1530 captured images was labeled and annotated using Roboflow. The labeling process was based on two classes: a track and a QR code. The labeled dataset was trained on a yolov8n-seg model - a small weight segmentation model by YOLO. Google CoLab was utilized because of its Hardware accelerator and GPU options for runtime environments. This decision helped speed up the training process to less than 45 minutes. Afterward, trained model metrics were analyzed to evaluate model efficiency. The training process was successful, producing a robust model. The main deciding factor for the efficiency of the model was the segmentation loss in proportion to the number of epochs. An epoch refers to the single entire passing of training data through a Machine Learning algorithm. A low segmentation loss meant a more efficient model. Segmentation masks produced by the model after performing an inference were used to calculate the deviation and orientation of the track. Thus, the model's segmentation accuracy was crucial for the precise movements of the wheelchair. Other parameters were also important, like recall and precision of the model. Fig 1.1 shows these metrics plotted against the number of epochs. The last step after training the model was to export the model from a PyTorch version (.pt) to a TensorFlow-lite version (.tflite), which can run on a Coral USB accelerator.

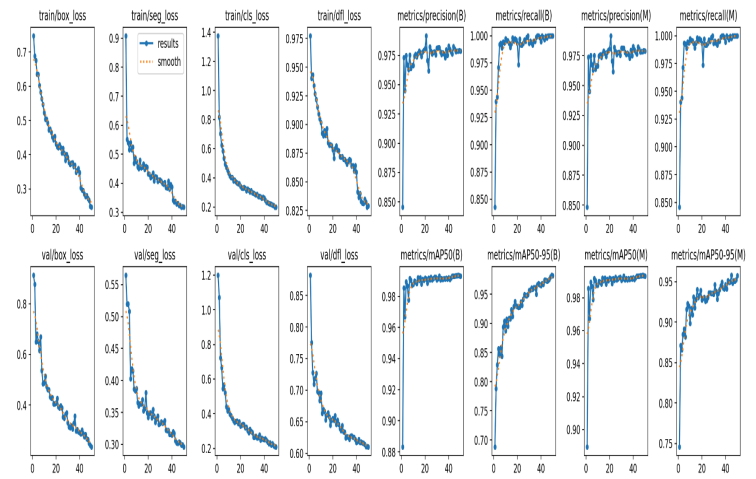


Fig 1.1 Different metrics of the model related to loss, precision, and recall plotted against the number of epochs.

After the model was trained and exported to TensorFlow-lite for the edge Tensor Processing Unit (TPU), an interface was designed for the model to detect results and properly handle and process those results. The whole coding process, from training to inference interface programming, was done in Python. OpenCV and Ultralytics libraries were utilized to perform classification, detection, and segmentation tasks based on a custom-trained model on yolov8n-seg. OpenCV provided APIs to access hardware resources, including the camera and graphics, and tools to perform frame handling and processing. The Ultralytics frameworks provided APIs to utilize the model and provide detection, classification, and segmentation results. Lastly, the Edge-tpu-silva library was utilized to allow the program to run on the Coral USB accelerator. Various classes were programmed to perform segmentation and detection, QR-code reading and handling, and deviation and orientation calculations.

A. Engineering Standards and Constraints

The implementation of Visual and LiDAR SLAM in an Autonomous Wheelchair is subject to a wide array of engineering standards and constraints. These ensure that the result is safe, reliable, and efficient. It follows safety standards like ISO 26262 to achieve functional safety for electrical and electronic systems, so fail-safe mechanisms will be available and the redundancy in critical components is ensured. Performance standards are based on real-time data processing, high accuracy in localization and mapping, and constant performance across environments: indoor and outdoor conditions under different lighting and surface conditions. The environmental constraints will mean that the sensor shall work in various conditions including variations in lighting, weather, surface types, and changes in temperature and humidity levels, which are typical for most indoor conditions found at airports.

Regulatory constraints provide that the wheelchair should be functional to all, including adherence to standards in accessibility like the Americans with Disabilities Act. Technical constraints relate to mounting visual and LiDAR sensors to gain better mapping and localization accuracy, computation load management for real-time processing without exorbitant power consumption, and power usage optimization for maximum battery life. Ethical and social constraints focus on the preservation of the privacy of the user, in which data that sensors gather must not invade the user's or other surrounding people's privacy. Safety is another constraint where the user must be safeguarded in every action or decision that is made by the smart environment.

Communication standards would include reliable wireless communication to aid remote monitoring and control. This could be followed by the use of IEEE 802.11 to standardize Wi-Fi or Bluetooth and strong security protocols in place to protect data. Mapping and localization constraints require detailed, high-resolution maps for accurate navigation in complex environments such as an airport, and dealing with the inability of GPS to work well indoors. SLAM relies much on visual and LiDAR data to enable very accurate localization. Software and algorithm standards underline the incorporation of best practices within the ROS 2 ecosystem in developing SLAM algorithms, ensuring these algorithms handle sensor noise, dynamic obstacles, and changes in the environment. It has to do with user interface and experience constraints, which are related to the interfaces needed to provide a feel of the control interface to users in an intuitive way, together with feedback mechanisms which might inform the user about the wheelchair status and required actions.

Specific considerations that can be identified in visual and LiDAR SLAM on a wheelchair are careful sensor installation to ensure the optimum field of view with no blind spots, frequent calibration of the sensors to ensure accuracy, effective data fusion algorithms for the proper use of each sensor's power, and sophisticated obstacle detection and avoidance algorithms to guarantee real-time response against both static and dynamic obstacles. Conforming to these rigorous engineering standards and constraints assures that the autonomous wheelchair system is safe, reliable, and effective for its intended use, providing a safe assistive technology for persons with mobility challenges. Leveraging Open-source standards and best practices, students will apply different kinds of standardized knowledge, and in many cases even combinations of these different types of standardized knowledge.

For the track implementation, most of the aforementioned engineering constraints are applied. Furthermore, the Track method requires track strips on the ground, which the wheelchair can follow, to work. This would mean the airport would have to reserve dedicated paths for the wheelchair, confining the wheelchair's mobility only to those places with

tracks.

The track poses advantages in lower processing and computing costs, yet it requires the physical restructuring of its environment to accommodate the tracks. Visual constraints were combated by using deep learning techniques on top of the traditional OpenCV computer vision techniques. The wheelchair has a LED system to illuminate its path when necessary, which also came with the aforementioned constraint of power consumption and utilization.

III. ANALYSIS

SLAM:

The integration of LiDAR and Visual SLAM components demonstrated the effectiveness of using a hybrid approach for autonomous navigation. The LiDAR SLAM provided robust 2D mapping and localization, essential for navigating flat surfaces and avoiding obstacles, while Visual SLAM offered detailed 3D environmental mapping and precise pose estimation, crucial for understanding complex surroundings and making informed navigation decisions. Through our analysis, we learned that combining multiple sensor modalities significantly enhances the system's overall performance and reliability. Accurate calibration and noise mitigation are critical for maintaining high-quality sensor data and improving SLAM outcomes. The Isaac ROS Visual SLAM framework, when properly configured and tuned, can deliver exceptional results in terms of mapping and localization accuracy. The insights gained from this project lay the groundwork for further improvements and potential real-world applications of autonomous wheelchairs in dynamic and complex environments like airports.

Our findings showed that the LiDAR SLAM system produced highly accurate 2D maps, with deviations from the ground truth typically within a few centimeters, consistent across multiple runs. Localization errors were generally small, with average deviations of less than 5 cm, demonstrating reliable performance in both static and dynamic environments. The LiDAR sensor exhibited low noise levels in static conditions; however, occasional spikes in measurement errors were observed in dynamic environments, highlighting the need for robust filtering techniques. We also found that the RealSense camera, combined with Isaac ROS Visual SLAM, generated dense and accurate 3D point clouds, closely matching the physical layout of the test environment with minor gaps in regions with low texture. Pose estimates were highly accurate, with average deviations from the ground truth trajectory within 3-4 cm. Disabling IMU fusion significantly reduced noise in the data, leading to more stable and reliable pose estimates.

Track:

Training the model on 50 epochs helped improve model efficiency. Based on training results, the segmentation loss

drastically went down after 40 epochs, which means if the model had been trained on less than 40 epochs, it would have been less efficient. Training the model on a lighter YOLO model also helped speed up the model on inference since the implementation worked with constricted hardware resources.

The use of the Google Coral USB accelerator helped speed up the model's performance. The model was able to process 22 to 30 Frames per second with the USB accelerator, which drastically increased processing speeds. Efficient algorithm design and frame handling also helped to ensure optimum speeds. Examples include resizing the frame and ensuring effective use of software resources within the program by utilizing Object Oriented Design Principles. The accuracy of the segmentation mask ensured precise calculations of deviation and orientation of the wheelchair. Running the program on the Raspberry Pi terminal also resulted in faster speeds with less load on the processor and the RAM.

IV. CONCLUSIONS

This paper presents the development and implementation of an advanced autonomous wheelchair system, leveraging SLAM-based models and track-based navigation to navigate complex environments such as airports. Our work addresses the significant challenges faced by existing semi-autonomous wheelchairs, particularly in achieving fully autonomous navigation through dynamic and obstacle-rich environments.

Our approach, developing methods for LiDAR and Visual SLAM, demonstrates a promising set of solutions for achieving accurate and reliable autonomous navigation. The LiDAR SLAM provided robust 2D mapping and localization, essential for avoiding obstacles and navigating flat surfaces, while the Visual SLAM offered detailed 3D environmental mapping and precise pose estimation, crucial for understanding complex surroundings. The Isaac ROS Visual SLAM framework, when properly configured and tuned, proved to deliver exceptional results in terms of mapping and localization accuracy, making it suitable for real-world applications in dynamic environments like airports. The track-based method, combined with deep learning and computer vision techniques, showed that a lower computational load could be achieved by physically restructuring the environment to accommodate tracks. This approach demonstrated efficiency in processing speeds and provided a reliable navigation path for the wheelchair, although it required significant infrastructure changes.

Several critical insights and recommendations emerged from our analysis. First, combining multiple sensor modalities significantly enhances the system's overall performance and reliability. Accurate calibration and noise mitigation are crucial for maintaining high-quality sensor data and improving SLAM outcomes. Second, efficient algorithm design, such as

resizing frames and utilizing object-oriented design principles, can drastically improve processing speeds and resource utilization, especially when working with constrained hardware. Third, our findings lay the groundwork for further improvements and potential real-world applications of autonomous wheelchairs in dynamic and complex environments. Continuous development and testing in varied settings will be essential to refine the system and ensure its robustness and reliability. Fourth, while the track-based method offers advantages in processing efficiency, it requires significant changes to the environment's physical layout. Future research venues will explore ways to integrate track-based navigation with minimal infrastructure modifications or develop adaptive systems that can transition seamlessly between track-based and SLAM-based navigation. Finally, ensuring user privacy and safety is paramount. Adhering to standards such as the Americans with Disabilities Act and implementing robust safety protocols and data privacy measures will be critical for the widespread adoption of autonomous wheelchairs.

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